University of Amsterdam
Conservation and Restoration
of Cultural Heritage

Workshop Report_Rev.01

UvA Master Programme Conservation and Restoration of Cultural Heritage

Name of student (student number): Moje Ryoo (13837486)
Specialisation: Contemporary Art
Course: **Conservation Principles and Practice 3: Contemporary Art**
Submission Date: 1st of July, 2024
*All the photos were taken by the student, unless stated otherwise.

[Light Systems & Contemporary Art]

Workshop 1: Activity log

- Date/ location: 11th.Apr. 2024 (09:30~17:00) / BSH F2.14, AG Contamporary art studio
- Lecturer : Evelyne Snijders
- Aim: Introduction about the light systems used in the contemporary art field / case study/ checking the suitable light bulb for the artworks in AG art studio
- Material preparation: Various type of light bulbs, electric plug, *Variation of chess match* by Berend Peter (Artwork)
- Checking process for the light bulb in the artwork:
  1) Select LED lights for replacement: 2700~4000 K / 8W / dimmable LED
  2) Connect the electric wire of the wooden table with a plug
  3) Check the light temperature and brightness of the original light bulb
  4) Check the light temperature and brightness of the LED lights
  5) Check the temperature of the light bulbs with heat gun
  6) Check the reflection and shadow of the inside panel with the light: white PMMA, black PMMA, wood, and cardboard

Workshop 1: Note and Result

- Kinds of light bulb: Incandescent, Halogen, Fluorescent, LED, Neon Type.
- For artwork that includes bulbs, it is recommended to change LED type because LED exerts less heat and has a longer life. However, in case the original light bulb has an aesthetic value itself, it needs to be preserved as much as possible.
- Incandescent bulb exerted heat immediately, it reached 150 C degrees in 20 seconds.
- LED bulb base also can exert considerable heat.
- **Suitable light bulb for the artwork: E27, 300-450 Lumen, 2500-2700K, milky white LED.**
- Inside panel material: most likely white PMMA, cardboard. In the future, white cardboard would be the first choice for the experiment in full shape.

Workshop 1: Reflection

There are several factors to be considered when dealing with light-involved artwork. Conservators need to examine the effect of heat on the artwork, as well as the aesthetic aspect. It also could be an ethical decision to replace the light bulb in a way, because of replacing the original part of the artwork, however, I learned that functionality and sustainability would be the priority in some cases. Since I have my artwork with an incandescent light bulb inside, I would like to research more about the effect of heat on PMMA sheets in the long term and find the proper LED light bulb as a replacement.

Workshop 1: Images

Fig.1. The original light bulb of the artwork / checking reflection with cardboard and white PMMA

Fig.2. LED: Relatively low heat / too white light temp. 2900K / the most similar color **2500-2700K**

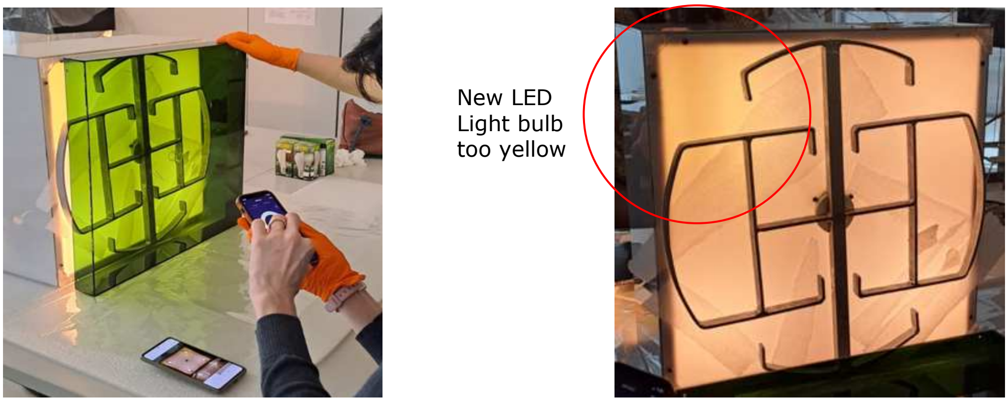


Fig.3. Other artwork/ checking brightness and reflection of colored PMMA with dimmerble LED bulb

[Electrical Systems]

Workshop 2: Activity log

- Date/ location : 12th.Apr. 2024 (10:00~17:00) / BSH F2.08B
- Lecturer : Paul Jansen Klomp (artist, teacher @ Leiden University Media Technology MSc program. https://uva.paulk.nl/)
- Aim: Understanding of electric system, multimeter and soldering
- Material preparation:
  -Multiple checker, soldering machine, soldering wire, breadboard.
  -Experiment5: Board, resistors, LED lights, capacitors, Integrated circuit, IC-socket, connector, 9V Battery, Headphone
- Process:
  1) Experiment1: Connect 10 Ohms resistor to a 9V battery
  2) Experiment2: Make series connection of 1K and 470E resistors on the breadboard
  3) Experiment3: Charge a 100uF capacitor using 9V battery and 1K resistor to flash LED
  4) Experiment4: Soldering with pre-soldering
  5) Experiment5: Make signal detector by instruction

Workshop 2: Note and Result

- Test results
  1) Experiment1: the resistor was bunt immediately with smell and smoke
     -10 Ohm resister cannot stand, 1000 Ohm would be OK.
  2) Experiment2: there was electricity flowing detected using a miltimeter.
     -Two resistors divide and flow the whole volts in the electricity.
  3) Experiment3: LED flashed, the capacitor can store the electricity.
  4) Experiment4: Electric wire was strongly connected by soldering.
     -Pre-soldering: some of soldering wire have resin inside, which is melted around 155C. It can be reducer, accelerating metal oxidation.
     Pre-soldering is good idea to coat the electric wire first.
     -It is important to keep the tip clean during soldering
     -Melting temp. of lead tip : 370C / non-lead tip: 400C
     -Difference between glue and soldering
     : Glue makes mechanical connection, soldering makes mechanical and electric connection.
     -It is important to stay still until the soldering wire is solidified. It affects electrical connection and the surface of soldering will have mat texture if moved.
  5) Experiment5: Succeeded. Refer the design sheet in the appendix.
     -IC socket is soldered instead of IC, to change the part in the future.
     -Each resistor makes stable electric current in the system.

Workshop 2: Reflection

By understanding electrical circuits and systems with theories, I thought I would be able to solve physically visible problems such as smoke or detachment of some soldering when small problems occur in artwork using electricity. However, calculating ohms or ampere is still hard for me, I think I can use some help from colleagues to confirm the exact result if necessary.

Workshop 2: Images

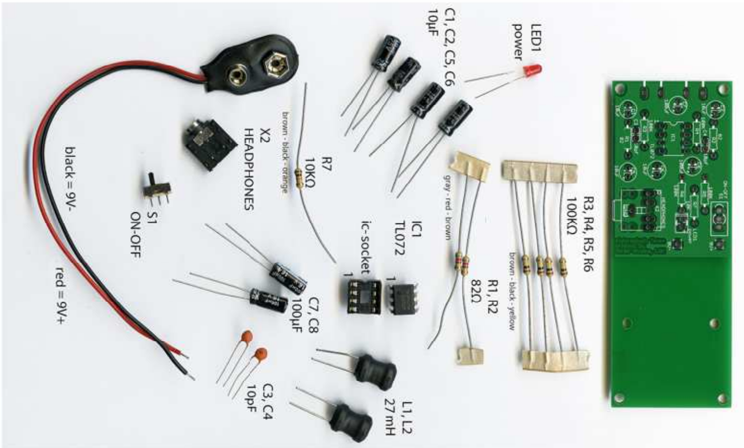
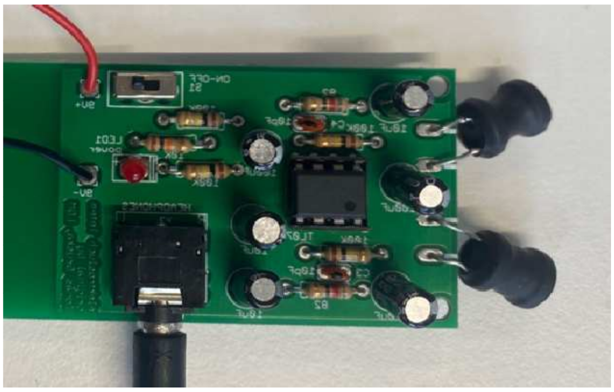


Fig.1. Material preparation for the experiments



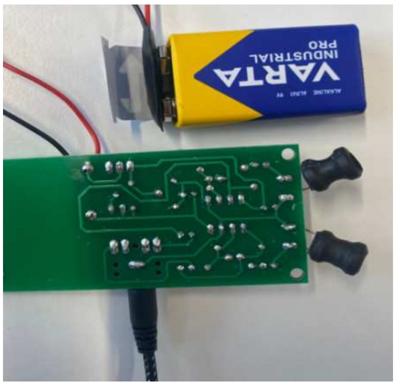
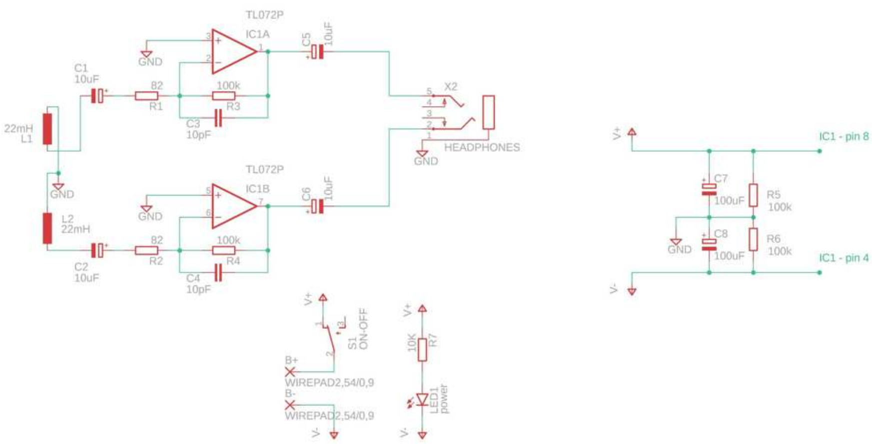


Fig.2. Experiment 5: Signal detector completed

Workshop 2: Appendix (source from https://uva.paulk.nl/)



[Computer-based and Software-based Art]

Workshop 3: Activity log

- Date/ location: 18th.Apr. 2024 (09:00~12:00) / Online
- Lecturer: Claudia Roeck (Time Based Media conservator at LIMA Amsterdam, HEK Basel)
- Aim: Computer and software based art conservation
- Case study1 : TraceNoizer (2001-2004)
  -LAN (Local Area Network): Annina Rüst, Fabian Thommen, Roman Abt, Silvan Zurbruegg and Marc Lee.
  -Artwork Link: http://www.tracenoizer.net/ (accessed on 1st of July.2024)
- Case study2 : Geert Mul, Shan Shui (2013)
  -Documentation video : https://vimeo.com/59690998 (accessed on 1st of July.2024)

Workshop 3: Note and Result

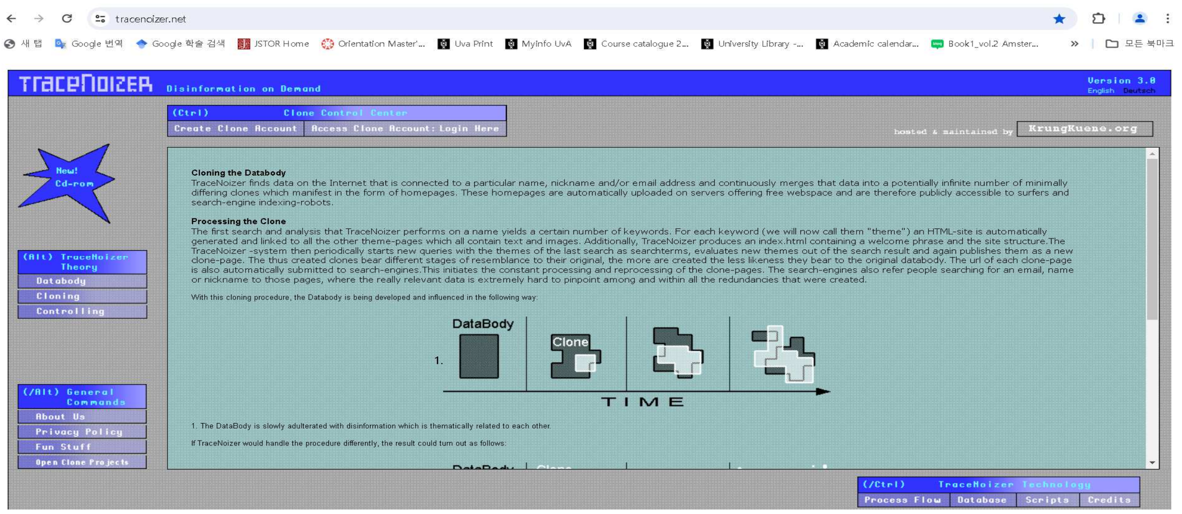
1) Characteristics of Software Based Art(SBA)
-In non-standard ecosystem
-Media player, computer are required
- Composition: Client(user, computer), web server, website
-Multiplicity of SBA: Acquired version ->Stabilized version supplied by artist
-> Permanent display version
-> Loans version
->Stabilized version made by museum staff
-> Cloned version for researchers
-Materiality of SBA: Information is material of artwork. Digital structure can resist their will, as much as clay, wood, or stone.
2) Conservation strategy
-Platform specific layer and platform independent layer artwork requires different strategy.
-Encapsulation: freezing environment and make a bridge to the new environment. It requires another layer to protect artwork or emulation.
-Migration: manually adapts new hardware. Repeated action is required.
-Reconstruction / Simulation: for some part of damaged artwork, cannot be reversible.
-Documentation as the artwork: In case damage was done, but no further damage expected. It Just needs to be kept the status
-Reinterpretation of conceptual art or continuation of processual art require different strategy.
3) Documentation of SBA
-Version control system
-Wiki system (Link or source code can be generated in test system)

Workshop 3: Reflection

Through this workshop, I got to think about the characteristics of obsolete in Computer and Software-based art. During the workshop, the sub-link for first example artwork TraceNoizer (2001-

2004) did not work. Even though the artwork has an important meaning that still applies today, that individuals cannot disappear on the Internet regardless of their will, it is unknown why it has not been restored yet. This type of artwork requires ongoing management, so it is very important to know what resources are needed and implement them.

Workshop 3: Images



- Artwork is intended to show impossibility to delete someone's trace in the internet world, and how easy clone of information can be made.

Fig.1. Case study 1: TraceNoizer (2001-2004), http://www.tracenoizer.net/

- Case study 2: Geert Mul, Shan Shui (2013)
- Link to artist's website: https://geertmul.nl/projects/shan-shui/ (accessed on 1st of July.2024)
- iPres article:https://osf.io/y3gcu
- Documentation video : https://vimeo.com/59690998 (accessed on 1st of July.2024)

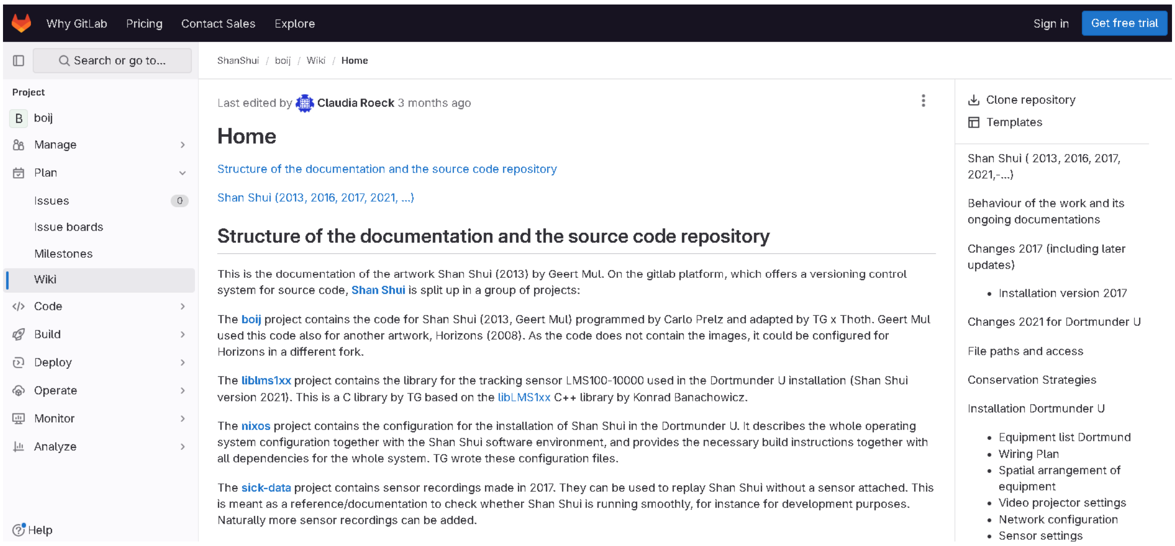


Fig.2. Example of documentation for SBA: Gitlab, version control system and Wiki.

[Studio visit Andreas Weisser]
Workshop 4: Activity log
• Date/ location: 19th.Apr. 2024 / JSC Düsseldorf, Schanzenstraße 54, 40549 Düsseldorf • Lecturer: Andreas Weisser and Julia Stoschek Foundation • Aim: visit studio/ archiving system of time based media collections Degradation of time bsed media hardware
Workshop 4: Note and Result
[Tour: Media storage] -Videotape, film, and slide storage located in the basement under a climate control system -Log system, and monitoring system (Model name OPUS 20) to keep 15C , 30-35 RH. Annual energy consumption is very high. 2 systems are acquired. If 1 is down, 1 immediately runs. -Rack system which magnetic properties were all removed / tested -Like fish, if new items come in, they need to be set in the temperature & RH in the storage. -there is a tunnel to adjust for 24 hours to take in a new acquired item. -Security camera is on 24/7, 2 cameras work independently to check they see the same scene. -Backup should be separate location (server) -Some of the artworks in hard drive and USB also come with play machines **[Video tapes]** -Very expensive in the early days -1 inch tape, -0.5 inch tape (Nam Jun Paik. Usually the pioneer of media artists used them) -U-matic : First Home video, consumer format, broadcasting industry use them. -VCR: home use, -VHS: 1990~2000s, -Video Cam (SONY): the most quality resolution, Digital base camera, -Digital cam, DV, mini DV -Razor disk (CD) : digital+analog+magnetic property -DVD: common data carrier, Hard drive -Wetransfer, but the cloud service on the internet is not suitable for archive artworks. -The question is: what tech the artwork relating to? What tech we can use on the artwork? **[Safe storage system]** -At least 3 location to be stored -Certificate by the official owner - Official contract : what to do & what not to do & long term preservation, right to show, loan. *"One copy is no copy" 3 copies are archived for the purpose of: 1) Original what we can get from the artist (not to be used.) 2) Same resolution, same quality (not to be used.) 3) Exhibition copy (to be used.) *What is important? -Presentation of original camera/file

-Presentation of content
-Is transcoding possible?
: if the old file format turned to be impossible to play, new system required.
If you can do, try to avoid transcode by yourself. It brings various risks.
*Discussion with the artist
-gathering of information with questions
-Manual, Certificate, Exhibition instruction, playback instructions and intentions
-Meta data is important most of the case
*Current issue of digital media preservation
-Making suitable database
-According to invention of the completely different version of programs and technologies,
museums are required to be equipped with VR, AI, etc. There early type of VR player is hard to find.

[Degradation of tape media]
1. Binder degradation
1) vinegar syndrome
-only effects cellulose-Acetate tapes, only audio tapes
-cause: aging of chemical/ bad quality original tape/ climate control failure
2)Hydrolysis / sticky-tape syndrome
-only affects polyester tape, audio & video tapes
-It remains residue on the machine, on the head of the machine
-If the tape is played in this status, it would be colorless -> blurred->blind.
-Tape cleaning: heat 50-55C , molecules will be rearranged. (The logic of tape cleaning machine)

2. Mechanical breakdown
-if mechanical problems, horizontal lines were shown up from to the bottom.
-cause: machine are misaligned
-video tape has separately stacked
-film was dislocated, not in the central location of the head and folded
-scratches on the film, tearing of tapes: apply special glue on the back side of the film.

3. Climate related problems
-mould, fungus, magnet layer,, especially in countries in humid conditions
-wavy structure: when the tapes cut and extended, with products from different manufacturers.
All the film can have different composition of chemicals, it creates different of level of wavy structure.

Workshop 4: Reflection

Listening to the lecture, I was curious about the extent of the conservator's domain. I thought about how the technical aspects would be managed in collaboration with experts, and which scope would be decided and executed directly by the conservator. Perhaps, conservators should take a more comprehensive view, as they manage not only the individual techniques applied to an artwork but also the budget and exhibition of the collection, and its preservation.

[Programming]

Workshop 5: Activity log

- Date/ location: 25-26th.Apr. 2024 (09:00~12:00) / BHS
- Lecturer: Reon Cordova (Artist and educator)
- Aim: Case studies and bilding interactive work on the internet, using Javascript
- Prior knowledge: based on my personal experience with html5 and CSS3
- Process: Day 1) learn how to use p5js.org and create personal interactive element
  < https://openprocessing.org/sketch/2252861>
  Day 2) make collective artwork: Dakoda-the-kitty
  <https://openprocessing.org/sketch/2252830>

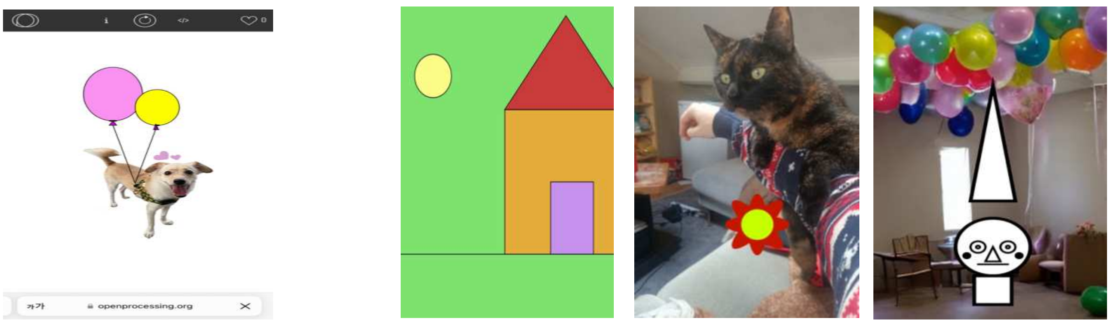
Workshop 5: Note and Result

- We leanred the same contents using p5 website, but everyone has a different way of applying codes to cearte visuals.
- Collective work expanded a lot of capability for the whole group. 'Divide and conquer' works.
- Experts will have simple and short code. The beginner's seems very long and inefficient.
- Logical organization of the codes and note are very critical factors for the preservation point of view.
- Programming means debugging in most cases.
- During the group project, ideas became abundant and we picked what can be done within the given time. We learned the time management is important in the project execution.

Workshop 5: Reflection

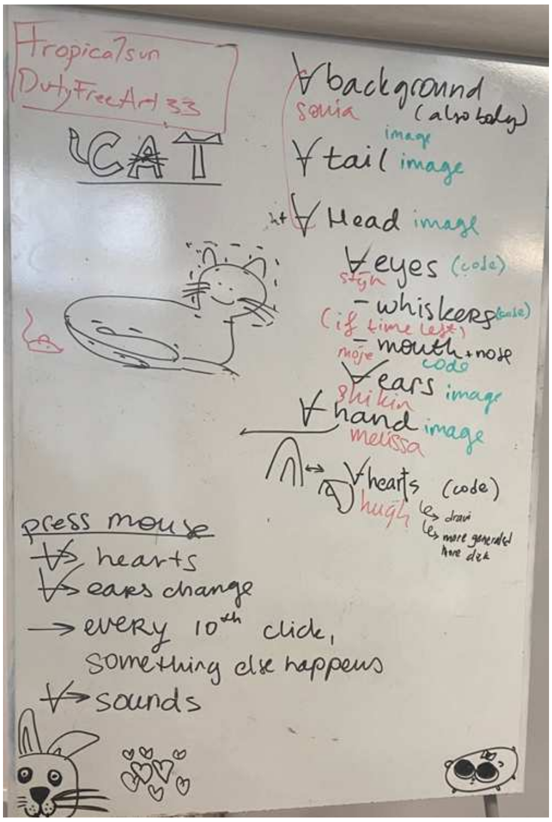
While programming, what I experienced was much more important than coding methods or skills, was the power of cooperation. In order to complete the whole task in time, we divided each part into parts and focused only on that part, achieving advanced results that surpassed each one's previous knowledge. I think we had a very valuable experience of teamwork by combining the parts we each completed and revising the finished work together.

Workshop5: Images



https://openprocessing.org/sketch/2252861 (accessed on 1st of July.2024)

Fig.1. Individual creation (and other student's creation)



Group project [Dakota-the-kitty]

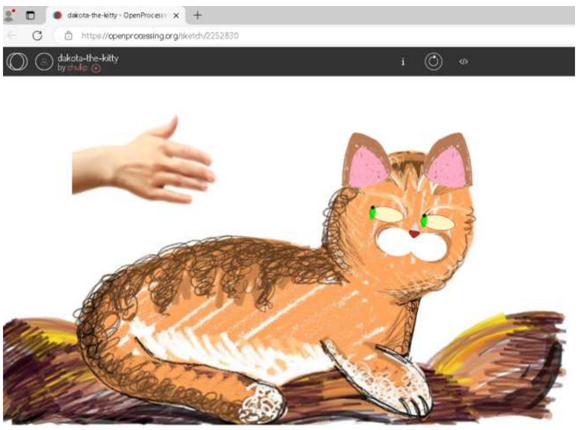
*Drawing and coding
-Background,tail,head: Sonia
-Eyes: Stijn
-Mouth and nose: Moje
-Ears: Shikin
-Hand: Melissa
-Hearts: Hugh

*Additions during prgress & applied
-Random jumpscare image
-Sounds

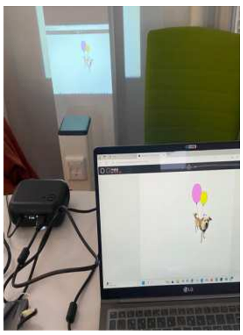
*Further ideas not applied
-An event happens by every 10 cliks
-Whiskers moving
-A tail moving

*Issue solved when publishing on the openprocessing.org webside: **Javascript library was changed.**

Fig.2. White board for our project : divide the sections, make each part and combine



https://openprocessing.org/sketch/2252830
(accessed on 1st of July.2024)



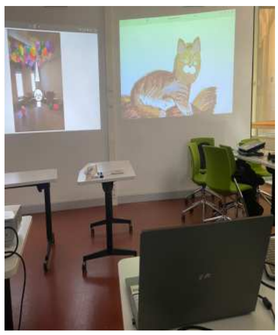
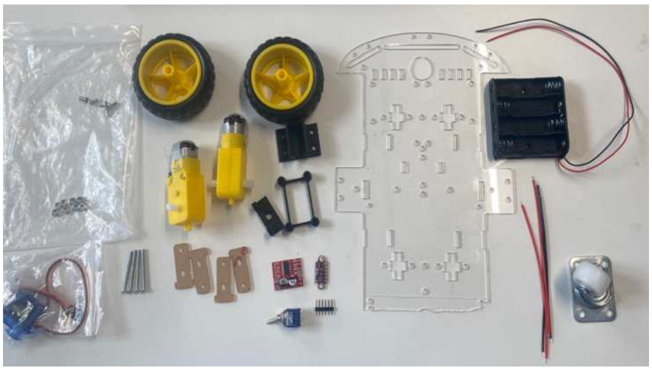


Fig.3. Final result of the group project and small exhibition photos

[Arduino workshop]

Workshop 6: Activity log

- Date/ location: 08th.May. 2024 (10:00~17:00) / BHS F2.08
- Lecturer: Paul Jansen Klomp (artist, teacher @ Leiden University Media Technology MSc program. https://uva.paulk.nl/)
- Aim: Understanding Arduino (UNO) in practice
- Material preparation:

- Process: 1) Assemble the hardware (Car)
  2) Load the software to the hardware
  * However, we were not able to load the software due to time limit.

Workshop6: Note and Result

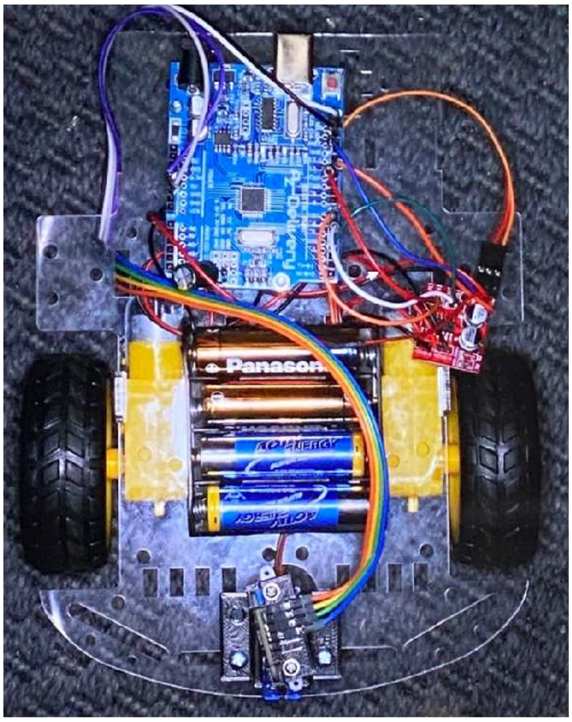
- Arduino is an open-source platform used for building electronics projects. Arduino consists of both a physical programmable circuit board (often referred to as a microcontroller) and a piece of software, or IDE (Integrated Development Environment) that runs on your computer, used to write and upload computer code to the physical board.[1]
- CPU : connected to power and ground
- CPU gains a number in a certain location in ROM, and gains instruction for the next step.
- CLK is connected to CPU, reading the beat
- ROM: Read-only memory, cannot be altered by CPU
- RAM: Random access memory, store calculations from CPU
- SSD: long term memory storage
- in/out connection of the analogue signals
- To execute Arduino, it requires 50-60 mm amp, power is enough with USB.
- Using LED blinking, it can find the error spot(debugging point)
- PWM: Pulse Width Modulation
- Servo motor : angle control, rotate 180 ~ 270 degree

Workshop 6: Reflection

[1] https://www.arduino.cc/en/Guide/Introduction (accessed on 1st of July.2024)

Assembling the Arduino RC car, I understood most of the procedures and executed them smoothly in general. However, because the DC motor's plus and minus were soldered in reverse, the wheel rolled backward when operated. Instead of re-soldering to solve the problem, I changed the position of the connected pin with Paul's help and the wheel rolled forward again. I learned that there are many ways to solve a problem if you follow the logic precisely.

Workshop6: Images



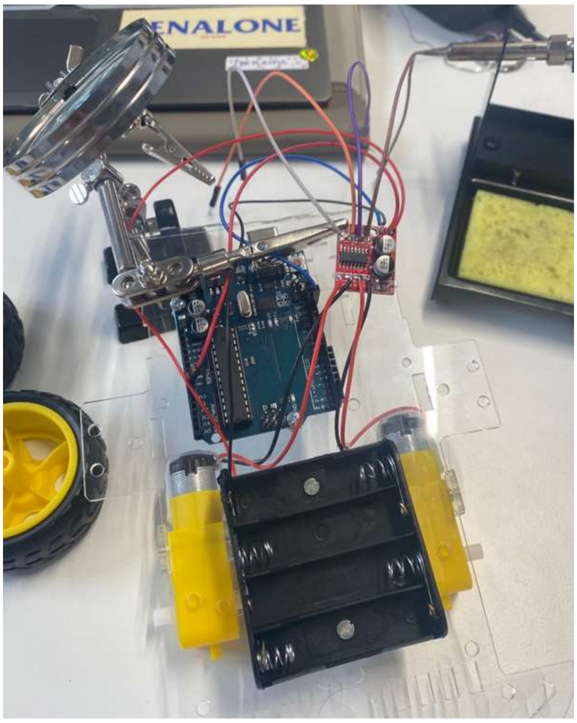


Fig.1. Assembly of hardware (example photo provided by the lecturer)

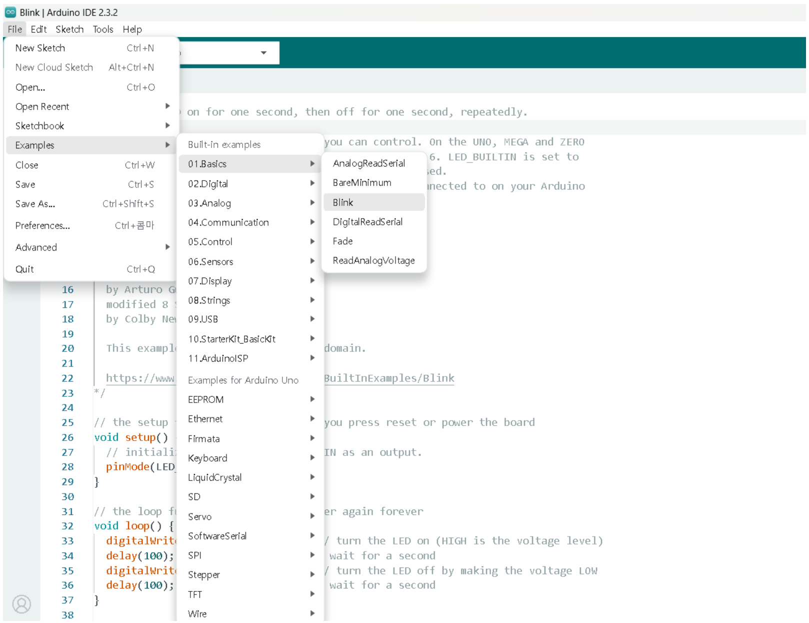


Fig.2. Arduino software example (download link: https://www.arduino.cc/)

[Raspberry Pi workshop]

Workshop 7: Activity log

- Date/ location: 18th.Apr. 2024 (09:00~12:00) / Online
- Lecturer: Paul Jansen Klomp (artist, teacher @ Leiden University Media Technology MSc program. https://uva.paulk.nl/)
- Aim: Understanding Raspberry Pi in practice
- Material preparation: 3 Raspberry Pi, 3 monitors, 3 mouses, 2 microSD cards(and adopters), a video clip, internet connection
- Process:
  - Day1 for playing a video clip, using Raspberry Pi
  1) Install Adafruit video player on 3 micro SD cards using Raspberry Pi imager program
  2) Put each card to 3 Raspberry Pi
  3) Connect Raspberry Pi to a monitor, a keyboard, and a mouse

  - Day2 for syncing a video clip on different monitors, using Raspberry Pi
  4) Install Mp4museum video player in 2 micro SD cards using Raspberry Pi imager
  5) Put each card to 2 Raspberry Pi in the same internet connection (router)
  6) Set each Raspberry Pi to sync its video clips, referring the manual
  : https://mp4museum.org/howto-sync/ (accessed on 1st of Jul.2024)
  7) Connect Raspberry Pi to a CRT monitor and Project beamer, a keybord, and a mouse
  8) Sync will be gradually accomplished when each Raspberry Pi starts playing

Workshop7: Note and Result

- Raspberry Pi is a single board computer for multi channel video player.
- Video player: Adafruit, mp4museum, etc.
- Use the interlace (to join different parts together to make a whole) method to play frames
- Compared to Arduino, Raspberry Pi has a video chip.
- Frame rate is important to show smooth video. 60 frames per second is standard in the EU, 30 is in the US. It affects video quality. If frame rate is not matched between a video and the player system, the system randomly picks frame and it makes lines or color changes on the screen.
- Double buffering method: screen & software work together to resolve the mismatches of frame rate. When the screen is reading /projecting, the software is writing/putting in the memory. It prepares the next frame safely.
- Most computers: every 60 frames per second, depending on graphic card/adjustment.
- One Raspberry Pi can be connected two monitors, but the analog video player can only have 1 monitor connection.
- Adafruit: video Looper program, can make playlist or random playing.
- Mp4museum: has a sync function and can make interactive videos. When plays, the two monitors gradually sync.

- Structure of blocs for data storage: partition system
- SONY CRT Monitor can handle 25P, Raspberry Pi changes it to HD PAL signal (50i) and display
- If the interlace was saved wrongly, the screen is jiggling or not smooth.
- Encording: 90~95% of original data will be missing when compressing images
- Ffmpeg : commend to computer about video disply. (in cmd in Window)
- DaVinci resolution: vedio editing program for the professionals in video industry. It can do editing of frame rate, switching filed orders, etc.

Workshop 7: Reflection

The debugging was a very important and time-consuming process. When we were all set for 2 Raspberry pis and Mp4museum software to make video sync between 2 computers, Raspberry pi could not locate the correct partition for the media file. I searched the website and found one saying "Windows might have trouble with showing media file's partition."[2] Eventually, Paul resolved issues and we learned that when issues happen, searching skills are important, using Github and searching information in the expert's group.

Workshop7: Images

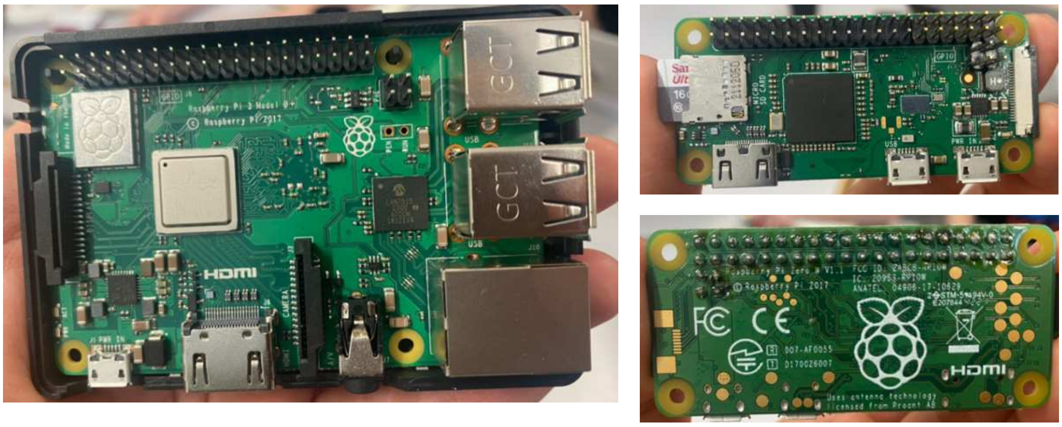


Fig.1. Raspberry Pi with different versions

Fig.2. Connections and sync videos in 2 monitors

2 https://mp4museum.org/howto/ (accessed on 1st of July.2024)

[Arduino assignment]

Assignment 1: Activity log

- Date/ location: 23th.May. 2024 (13:00~17:00) / BSH
- Assignment: Add RGB ring to RC car with Arduino, design codes to light the RGB ring
- Material preparation: Arduino IDE program / hardware(car)/USB cable/RGB ring (WS2812B 12-bit)/soldering machine/soldering wire/3 jump cables
- Process:
  1) Read the assignment instruction provided
  2) Download the Adafruit Neopixel library
  3) Apply 'simple' example code in Arduino IDE
  4) Apply various codes related to the RGB light ring in Arduino IDE
  5) Troubleshooting
  6) Record the final code and analysis

Assignment 1: Final code for RC Car

```
//----------------------------------------------------- Start
//----------------------------------------------------- Initialization

long duration;
//float forwarddistance;
//float distancefront;
float distanceArr[6];

#include <Servo.h>
#define pinServo 12
#include <Wire.h>
#include <VL53L0X.h>
VL53L0X sensor;

//----------------------------------------------------- Initialization: Setting up Servo
Angles
int servoArray[6] = { 0, 25, 50, 110, 145, 180 };
Servo myServo;
//----------------------------------------------------- Initialization: Motors
int motor_left[] = { 3, 5 };
int motor_right[] = { 6, 11 };
//----------------------------------------------------- Initialization: Setup Module
#include <Adafruit_NeoPixel.h>
#ifdef __AVR__
#include <avr/power.h>
#endif
#define PIN        9   //pin number RGB ring use
#define NUMPIXELS 12  //how many LED in the ring

Adafruit_NeoPixel pixels(NUMPIXELS, PIN, NEO_GRB + NEO_KHZ800);
#define DELAYVAL 300
#define DELAYVAL1000 1000
//----------------------------------------------------RGB ring - assignment
```

* Assignment: Install Adafruit-Neopixel library

```
void setup() {
#if defined(__AVR_ATtiny85__) && (F_CPU == 16000000)
  clock_prescale_set(clock_div_1);
#endif

  pixels.begin();
  //RGB ring
  pixels.setBrightness(10);


  Serial.begin(9600);
  myServo.attach(pinServo);
  //Setup Motors
  for (int i = 0; i < 2; i++) {
    pinMode(motor_left[i], OUTPUT);
    pinMode(motor_right[i], OUTPUT);
  }
  // setup VL53L0x distance meter
  Wire.begin();
  sensor.setTimeout(300);
  if (!sensor.init()) {
    Serial.println("Failed to detect and initialize sensor!");
    while (1) {}
  }
  // sensor.startContinuous();
}

//----------------------------------------------------- Initialization: Motor Module
// Following Module is for the Robot to brake
void brake() {
  // 1 1 , 1 1
  pixels.clear();
  digitalWrite(motor_left[0], HIGH);
  digitalWrite(motor_left[1], HIGH);
  digitalWrite(motor_right[0], HIGH);
  digitalWrite(motor_right[1], HIGH);
}
// Following Module is for the Robot to go forward


void drive_forward() {
  // 0 1 , 0 1
  pinMode(3, OUTPUT);
  analogWrite(3,100);// Mortor speed control
  pinMode(3, OUTPUT);
  digitalWrite(3, LOW);// Motor off

  pinMode(5, OUTPUT);
  analogWrite(5, 100);// Mortor speed control
  pinMode(5, OUTPUT);
  digitalWrite(5, HIGH);// Motor on

  pinMode(6, OUTPUT);
```

* Assignment: lights all off when brake

* Unsolved issue
: Please refer a note at the end

```
  analogWrite(6, 100);// Mortor speed control
  pinMode(6, OUTPUT);
  digitalWrite(6, LOW);// Motor off

  pinMode(11, OUTPUT);
  analogWrite(11, 100);  // Mortor speed control
  pinMode(11, OUTPUT);
  digitalWrite(11, HIGH);// Motor on


}
// Following Module is for the Robot to slightly go towards left
void slight_left() {
  // 0 1 , 0 0
  digitalWrite(motor_left[0], LOW);
  digitalWrite(motor_left[1], HIGH);
  digitalWrite(motor_right[0], LOW);
  digitalWrite(motor_right[1], LOW);
  delay(150);
}
// Following Module is for the Robot to slightly go towards right
void slight_right() {
  // 0 0 , 0 1
  digitalWrite(motor_left[0], LOW);
  digitalWrite(motor_left[1], LOW);
  digitalWrite(motor_right[0], LOW);
  digitalWrite(motor_right[1], HIGH);
  delay(150);
}
// Following Module is for the Robot to turn left
void turn_left() {
  // 0 1 , 0 0
  digitalWrite(motor_left[0], LOW);
  digitalWrite(motor_left[1], HIGH);
  digitalWrite(motor_right[0], LOW);
  digitalWrite(motor_right[1], LOW);
  delay(400);
}
// Following Module is for the Robot to turn right
void turn_right() {
  // 0 0 , 0 1
  digitalWrite(motor_left[0], LOW);
  digitalWrite(motor_left[1], LOW);
  digitalWrite(motor_right[0], LOW);
  digitalWrite(motor_right[1], HIGH);
  delay(400);
}
// Following Module is to make a uturn if there are obstacles to the right, to the left and
in front
void back_up() {
  // 1 0 , 1 0 eq reverse

      pixels.setPixelColor(0, pixels.Color(255,0,0));
      pixels.setPixelColor(1, pixels.Color(255,0,0));
```

*Assignment a. Backward, LED red
(setPixelColor can be declared in Void
or directly in else-if statement.)

```
      pixels.setPixelColor(2, pixels.Color(255,0,0));
      pixels.setPixelColor(3, pixels.Color(255,0,0));
      pixels.setPixelColor(4, pixels.Color(255,0,0));
      pixels.setPixelColor(5, pixels.Color(255,0,0));
      pixels.setPixelColor(6, pixels.Color(255,0,0));
      pixels.setPixelColor(7, pixels.Color(255,0,0));
      pixels.setPixelColor(8, pixels.Color(255,0,0));
      pixels.setPixelColor(9, pixels.Color(255,0,0));
      pixels.setPixelColor(10, pixels.Color(255,0,0));
      pixels.setPixelColor(11, pixels.Color(255,0,0));//when back, red
      pixels.show();

      digitalWrite(motor_left[0], HIGH);
      digitalWrite(motor_left[1], LOW);
      digitalWrite(motor_right[0], HIGH);
      digitalWrite(motor_right[1], LOW);
      delay(400);

}
//------------------------------------------------------- Sight Forward
float distance(int angle) {
  // returns actual reading if > 0 and < 400, or else returns 2000
  myServo.write(angle);
  int fwd = sensor.readRangeSingleMillimeters();
  if ((fwd > 0) and (fwd < 400))
    return (fwd);
  else
    return (2000);
}

//------------------------------------------------------- Sensor Array Module
void scan_around() {
  sensor.startContinuous();
  for (int i = 0; i < 6; i++) {
    myServo.write(servoArray[i]);
    if (i == 0)
      delay(300);
    else
      delay(100);
    int reading = sensor.readRangeContinuousMillimeters();
    if (reading > 0 && reading < 400)
      distanceArr[i] = reading;
    else
      distanceArr[i] = 2000;
    Serial.print("distanceArr[");
    Serial.print(i);
    Serial.print("] = ");
    Serial.println(distanceArr[i]);
  }
  sensor.stopContinuous();
  // reset servo to center position
  myServo.write(75);
  delay(300);
}
```

```
//--------------------------------------------------------- Loop Module

int thresh = 300;      // threshold for free space
int fwd_thresh = 400;  // threshold for reaction when driving forward

void loop() {
  pixels.clear();
  int distancefront = distance(75);

  Serial.println(distancefront);
  if (distancefront > fwd_thresh) {

    drive_forward();
    delay(100);
    Serial.println("1");
  } else {
    brake();
    pixels.clear();
    scan_around();
    if ((distanceArr[0] > 1000) && (distanceArr[1] > 1000) && (distanceArr[2] > 1000) &&
(distanceArr[3] > 1000) && (distanceArr[4] > 1000) && (distanceArr[5] > 1000)) {
      Serial.println("Restarted the Loop");
      delay(100);
      //return;  // hardly ever happens, all fields clear after finding an object while
driving forward: return -> restart the main loop

    } else if (((distanceArr[1]) > thresh) && ((distanceArr[2]) > thresh)) {
      pixels.setPixelColor(0, pixels.Color(0,255,0));
      pixels.setPixelColor(1, pixels.Color(0,255,0));
      pixels.setPixelColor(2, pixels.Color(0,255,0));
      pixels.setPixelColor(3, pixels.Color(0,255,0));
      pixels.setPixelColor(4, pixels.Color(0,255,0));
      pixels.setPixelColor(5, pixels.Color(0,255,0));
      pixels.setPixelColor(6, pixels.Color(0,255,0));
      pixels.setPixelColor(7, pixels.Color(0,255,0));
      pixels.setPixelColor(8, pixels.Color(0,255,0));
      pixels.setPixelColor(9, pixels.Color(0,255,0));
      pixels.setPixelColor(10, pixels.Color(0,255,0));
      pixels.setPixelColor(11, pixels.Color(0,255,0)); //when slight right turn, green
      pixels.show();
      slight_right();
      //pixels.clear();
      drive_forward();
      delay(100);
      Serial.println("2");

    } else if (((distanceArr[0]) > thresh) && ((distanceArr[1]) > thresh)) {
      pixels.setPixelColor(0, pixels.Color(0,0,255));
      pixels.setPixelColor(1, pixels.Color(0,0,255));
      pixels.setPixelColor(2, pixels.Color(0,0,255));
      pixels.setPixelColor(3, pixels.Color(0,0,255));
      pixels.setPixelColor(4, pixels.Color(0,0,255));
      pixels.setPixelColor(5, pixels.Color(0,0,255));
      pixels.setPixelColor(6, pixels.Color(0,0,255));
```

*Assignment b.
Slight right turn, LED green
(the location of LED can be selected)

*Assignment c.
Right turn, LED blue

```
    pixels.setPixelColor(7, pixels.Color(0,0,255));
    pixels.setPixelColor(8, pixels.Color(0,0,255));
    pixels.setPixelColor(9, pixels.Color(0,0,255));
    pixels.setPixelColor(10, pixels.Color(0,0,255));
    pixels.setPixelColor(11, pixels.Color(0,0,255));//when right turn, blue
    pixels.show();
    turn_right( );
    //pixels.clear();
    drive_forward();
    delay(100);
    Serial.println("3");

  } else if (((distanceArr[2]) > thresh) && ((distanceArr[3]) > thresh)) {
```

*Assignment d.
Slight left turn, LED magenta

```
    pixels.setPixelColor(0, pixels.Color(255,0,255));
    pixels.setPixelColor(1, pixels.Color(255,0,255));
    pixels.setPixelColor(2, pixels.Color(255,0,255));
    pixels.setPixelColor(3, pixels.Color(255,0,255));
    pixels.setPixelColor(4, pixels.Color(255,0,255));
    pixels.setPixelColor(5, pixels.Color(255,0,255));
    pixels.setPixelColor(6, pixels.Color(255,0,255));
    pixels.setPixelColor(7, pixels.Color(255,0,255));
    pixels.setPixelColor(8, pixels.Color(255,0,255));
    pixels.setPixelColor(9, pixels.Color(255,0,255));
    pixels.setPixelColor(10, pixels.Color(255,0,255));
    pixels.setPixelColor(11, pixels.Color(255,0,255));//when slight left turn, magenta
    pixels.show();
    slight_left();
    //pixels.clear();
    drive_forward();
    delay(100);
    Serial.println("4");

  } else if (((distanceArr[4]) > thresh) && ((distanceArr[5]) > thresh)) {
```

*Assignment e.
Left turn, LED yellow

```
    pixels.setPixelColor(0, pixels.Color(255,255,0));
    pixels.setPixelColor(1, pixels.Color(255,255,0));
    pixels.setPixelColor(2, pixels.Color(255,255,0));
    pixels.setPixelColor(3, pixels.Color(255,255,0));
    pixels.setPixelColor(4, pixels.Color(255,255,0));
    pixels.setPixelColor(5, pixels.Color(255,255,0));
    pixels.setPixelColor(6, pixels.Color(255,255,0));
    pixels.setPixelColor(7, pixels.Color(255,255,0));
    pixels.setPixelColor(8, pixels.Color(255,255,0));
    pixels.setPixelColor(9, pixels.Color(255,255,0));
    pixels.setPixelColor(10, pixels.Color(255,255,0));
    pixels.setPixelColor(11, pixels.Color(255,255,0));//when left turn, yellow
    pixels.show();
    turn_left();
    //pixels.clear();
    drive_forward();
    delay(100);
    Serial.println("5");
```

```
    } else {
      back_up();
      pixels.clear();
      brake();
      delay(1000);
      Serial.println("6");
      return;
    }}
```

```
  //-----RGB light during going straight

  for (int i=0; i<NUMPIXELS; i++) {

   pixels.setPixelColor(i,
pixels.Color(random(0,255),random(0,255),random(0,255)));//random color
   pixels.show();
   delay(DELAYVAL);

  }
```

*Assignment:lights all on when driving forward
random color
delay 300 for every single LED

```
 } //end
```

[* Unsolved issue]

When the RC car was turned on, it went straight fast and crushed the wall at the end. I presumed that the sensor was supposed to sense the wall before the car touched it, and I tried to slow the motor speed down. I searched the code and applied 'analogWrite()' in the motor control section. However, it did not work well.

I found that it was asked by many people online why 'analogWrite()' is not working properly with following 'digitalWrite()' when they are written together(screen capture in *Image*).[2] Until now, unfortunately, it **has not been clear** to me how I can control the DC motor speed only with codes.

Assignment 1: Imgages

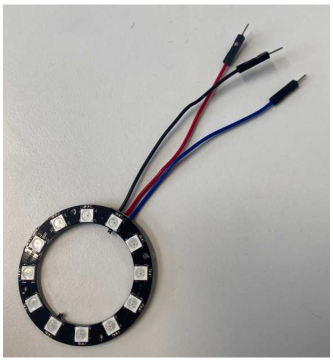
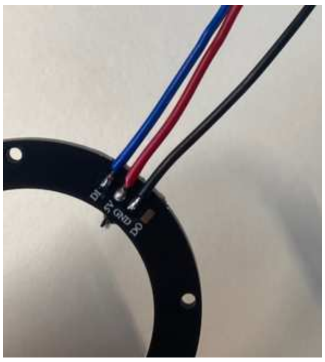



Fig.1. Connection and soldering of RGB Rings to Arduino board in RC car

[2] https://forum.pjrc.com/index.php?threads/digitalwrite-doesnt-work-after-analogwrite-on-the-same-pin.9/ (accessed on 1st of July.2024)

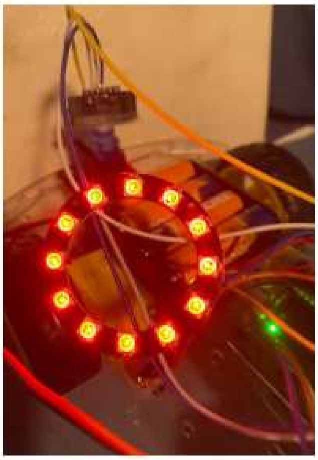
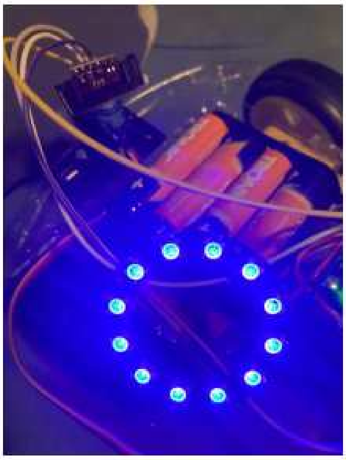
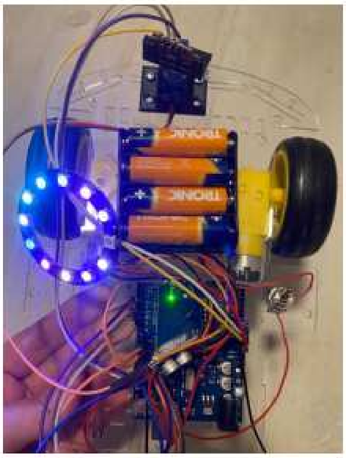
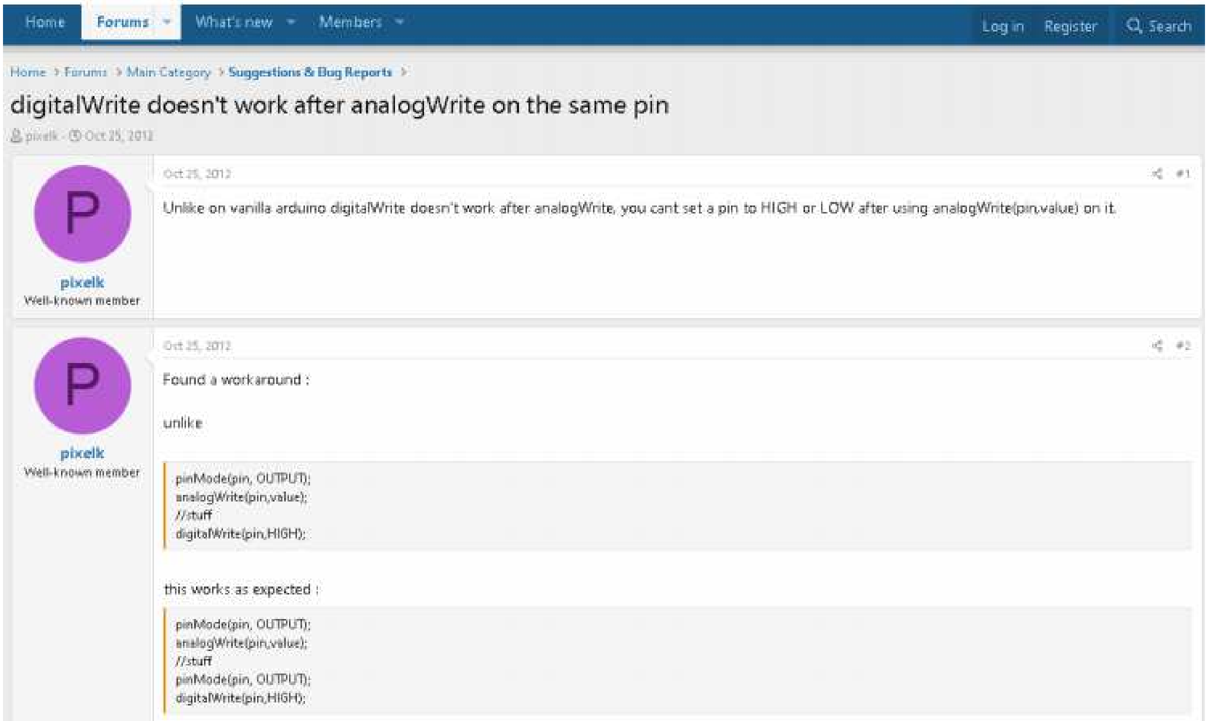
		
Back-red	Turn right-blue	Forward-random color/delay300

Fig.2 RGB Lights on by commands



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digitalWrite doesn't work after analogWrite on the same pin

pixelk · Oct 25, 2012

P pixelk Well-known member

Oct 25, 2012 #1

Unlike on vanilla arduino digitalWrite doesn't work after analogWrite, you cant set a pin to HIGH or LOW after using analogWrite(pin,value) on it.

P pixelk Well-known member

Oct 25, 2012 #2

Found a workaround :

unlike

```
pinMode(pin, OUTPUT);
analogWrite(pin,value);
//stuff
digitalWrite(pin,HIGH);
```

this works as expected :

```
pinMode(pin, OUTPUT);
analogWrite(pin,value);
//stuff
pinMode(pin, OUTPUT);
digitalWrite(pin,HIGH);
```

Fig.3 An unsolved issue on 'analogWrite' and 'digitalWrite codes

Assignment 1: Reflection

To execute the task, I searched and downloaded the Neopixel library from the Arduino IDE. Before modifying the final code completed in the lecture with Paul, I rechecked the pins where the DC motors are connected and the servo motor, which scans the section divided by the 6 angles (sevoArray). I also made sure that I wrote all the notes and that there were no problems with the operation. I had to solder to attach the RGB ring, but the space between the three cables was very narrow. So I needed to be careful not to connect the three, using a small amount of soldering wire because there is a risk of short circuits.

RGB Ring's GND is connected to GND on the Arduino board, DI is connected to an input (pin no.9), and 5V is connected to the power. Then, I specified colors in the code (pixels.setPixelColor), where the sensor reads the distance to the obstacle in front and determines the bending angle so that the code generates a different color when the car changes direction. Finally, it worked well except for the unsolved issue mentioned above in the assignment report.

Another problem I faced during the practice was that while checking the color changing of the RGB ring, a soldering of the cable connected to the servo sensor and Arduino board was broken. I had to go to Bushuis to have it re-soldered. The sensor had to move a lot to read the angle, so it broke a few times during practice. After 4 times of re-soldering, I found that if I did the pre-soldering from the beginning, it would last longer under the repeated movement. If I do soldering in the future, it will be important to make it strong enough, using the pre-soldering method.

In the future, if I have a chance to handle or conserve the artwork using Arduino, I could understand the basics of it and do the physical examination. As I searched, there is a Korean media artist group creating artworks using Arduino in Youtube[3], I would like to take a look and learn from their creations.

[3] https://www.youtube.com/@LOVOTLAB (accessed on 1st of July.2024)